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Theoretical modelling and prediction of surface roughness for hybrid additive–subtractive manufacturing processes

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ABSTRACT

Hybrid additive–subtractive manufacturing processes are becoming increasingly popular as a promising solution to overcome the current limitations of Additive Manufacturing (AM) technology and improve the dimensional accuracy and surface quality of parts. Surface roughness, as one of the most important surface quality measures, plays a key role in the fit of assemblies and thus needs to be thoroughly evaluated at the design and manufacturing stages. However, most of the studies on surface roughness modelling and analysis employ empirical approaches, and only consider the effect of a single manufacturing process. In particular, the existing surface roughness models are not applicable to hybrid additive–subtractive manufacturing processes in which a secondary process is involved. In this article, analytical models are established to predict the surface roughness of parts fabricated by AM as well as hybrid additive–subtractive manufacturing processes. A novel surface profile representation scheme is also proposed to increase the prediction accuracy. Case studies are performed to validate the effectiveness of the proposed models. An average of 4.25% error is observed for the AM case, which is significantly smaller than the prediction error of the existing models in the literature. Furthermore, in the hybrid case, an average of 91.83% accuracy is obtained.

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Additive manufacturing; hybrid additive–subtractive manufacturing process; milling process; surface roughness; surface profile

1. Introduction

Additive Manufacturing (AM) is a novel manufacturing technique that directly fabricates parts using a layer-wise approach and a three-dimensional (3D) digital model (ASTM, 2012). Compared with traditional manufacturing processes, AM technology offers higher levels of customization and flexibility in manufacturing complex structures and functionally graded materials, and has great potential in reducing material waste and manufacturing cost (Hopkinson *et al.*, 2006; Gao *et al.*, 2015). Consequently, numerous applications of AM in different industries including aerospace, automotive, and healthcare have been investigated and practiced by researchers over the past two decades (Guo and Leu, 2013).

The control and prediction of the obtained quality characteristics in different manufacturing processes, including AM, is a critical requirement for designers and manufacturers (Feng and Wang, 2003) that not only allows for checking the compliance with design specifications, but also helps with the determination of manufacturing strategies in the process planning stage (Geddam and Kaldor, 1998). Therefore, extensive research efforts have been dedicated to the characterization, control, and prediction of dimensional, geometrical, and surface properties in different AM processes through both empirical and analytical means. As an example, different quality measures, such as dimensional accuracy (Sood *et al.*, 2009; Huang *et al.*, 2015; Haghighi et al. 2017; Cheng *et al.* 2018; Haghighi and Li, 2018), geometrical errors (Arni and Gupta, 2001; Hanumaiah and Ravi 2007; Ollison and Berisso, 2010; Paul and Anand, 2011; Martorelli *et al.*, 2017), porosity (Wu *et al.*, 2018), and surface properties (Santo, 2014) have been investigated in the literature.

Surface quality is one of the most important quality measures, and is mainly evaluated by the surface roughness parameters defined by the ISO standard (ISO, 1997), which depends on several factors including material properties and process planning parameters. Despite the numerous advantages of AM technology, the surface quality of printed parts is generally inferior to those fabricated by traditional subtractive processes using Computer Numerical Control machines (Wong and Hernandez, 2012; Krolczyk *et al.*, 2014). This relatively lower surface quality is mainly caused by the stair-stepping error due to the AM's layer-by-layer production method, and is significantly influenced by process parameters such as layer thickness and build orientation (Vasudevarao *et al.*, 2000; Pandey *et al.*, 2007; Taufik and Jain, 2013). Furthermore, the support generation and removal that are required in most AM processes often result in additional deformation and quality issues, especially if not performed properly. Consequently, extensive research has been dedicated to surface quality improvement through process parameter optimization, which is mainly performed using an empirical approach. However, the surface quality improvements are still not significant enough to comply with the surface quality standards of subtractive processes, and may not be satisfactory for meeting design requirements, especially in the case of tight tolerances. An innovative approach for improving the surface quality of AM parts is by leveraging the advantages of both additive and subtractive processes through the

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development of hybrid additive–subtractive manufacturing processes (Karunakaran *et al*., 2010; Flynn *et al*., 2016). In the majority of the developed hybrid additive–subtractive manufacturing processes, the subtractive process is solely employed for surface finishing, performed to improve the quality of printed surfaces, and is thus considered as a secondary process (Pandey *et al*., 2003; Boschetto *et al*., 2016).

To assess the compliance with the design specifications, *a priori* prediction of surface properties (e.g., surface roughness) as a result of the hybrid additive–subtractive manufacturing process is necessary. Although various surface roughness modelling studies are reported in the literature for different AM processes, analysis of surface roughness in hybrid additive–subtractive manufacturing processes is rarely addressed. In other words, the existing surface roughness models for AM processes rely on the assumption that no secondary process (e.g., milling or turning) is performed. Furthermore, most surface roughness modelling studies are based on empirical approaches, and are usually limited to a certain AM process, e.g., the Fused Deposition Modelling (FDM) process. Therefore, the objective of this article is to establish comprehensive mathematical models for the prediction of surface roughness in hybrid additive–subtractive manufacturing processes. The developed models are specifically tailored to hybrid processes in which a flat end mill tool is utilized as the subtractive process for milling the printed surfaces. In addition, an accompanying algorithm is developed and used to obtain the unknown parameters in the proposed models. Furthermore, a novel surface profile representation scheme for 3D printed surfaces is proposed to increase the prediction accuracy compared with the existing models in the literature.

The rest of this article is organized as follows. Section 2 provides an overview of the literature on surface roughness analysis and prediction in different AM processes. A novel surface profile representation scheme for printed surfaces in AM process is proposed in Section 3 and used to establish the surface roughness model for AM process. Section 4 presents the developed mathematical models for predicting the surface roughness in hybrid additive–subtractive manufacturing processes. The proposed algorithm for estimating the unknown parameters of the developed models is presented in Section 5. The numerical results from the case studies for the purpose of model validation and comparison with literature are reported in Section 6. Finally, the conclusions and future work are discussed in Section 7.

2. Literature review

Analysis of the obtained surface roughness in AM processes is of great interest to researchers. Empirical approaches form the majority of studies in the literature and provide useful guidance for understanding the effect of process parameters on the obtained surface roughness values. A semi-empirical study was performed by Chryssolouris *et al*. (1999) in which the correlation of process parameters and surface roughness was statistically studied for the Laminated Object Manufacturing process. A fractional factorial design of experiments was applied in Vasudevarao *et al*. (2000) to study the effects of layer thickness, build orientation, road width, air gap and temperature on the surface finish in FDM process. Anitha *et al*. (2001) used the Taguchi technique to evaluate the effects of process parameters on surface roughness in FDM process, where a strong inverse relationship between layer thickness and surface roughness was observed. Campbell *et al*. (2002) developed a methodology and an accompanying software application tool to visualize the surface roughness value in different build orientations, and guide designers to select the optimum build orientation. The different error sources in FDM process were identified and quantified in terms of surface roughness, dimensional accuracy, and precision in Bochmann *et al*. (2015). Recently, an adaptive neuro-fuzzy inference system was developed in Dambatta and Sarhan (2016) for the prediction of the surface roughness in FDM processes; it was based on critical process parameters, namely build orientation and layer thickness. The above empirical studies highlight the significant effect of layer thickness and build orientation (also referred to as stratification angle, build angle or surface angle) on the surface roughness. Furthermore, according to the results of these studies, surface roughness is observed to behave differently for different ranges of stratification angle.

Despite the necessity of empirical approaches for providing preliminary insights on the effect of different process parameters, comprehensive analytical models are generally preferred, but less addressed in the literature, due to the increased level of complexity. Luis Perez *et al*. (2001) were among the first researchers that addressed this issue, and established an analytical model for surface roughness using two different surface profile representation schemes, i.e., sharp edge and round edge. A prediction model for surface roughness in FDM process was proposed by Ahn *et al*. (2009) where the inclination of the surface profile was also considered in addition to the layer thickness and stratification angle parameters. Assuming that the surface profile has a slightly platykurtic distribution, Boschetto *et al*. (2012) proposed a periodic arc-based shape to approximate the surface profile in FDM process. A good agreement between the theoretical model and the experimental data was observed for surface angles between 30° and 150°. Strano *et al*. (2013) further extended the existing surface roughness models for FDM process to the Selective Laser Melting process. They managed to decrease the prediction error by incorporating the presence of particles on the top surfaces into the model. More recently, a new modelling methodology for the FDM profile was proposed by Taufik and Jain (2016) that categorizes three types of build edge profiles: perimeter, raster, and combination of both patterns. The authors also proposed a periodic parabolic-based profile scheme to represent the surface profile. Boschetto and Bottini (2015) were among the first researchers to address the surface roughness modelling in hybrid FDM and Barrel Finishing (BF) process. The authors incorporated two parameters of layer thickness and deposition angle from the AM process, and material removal from the BF process into their model. However, their model did not address other subtractive processes such as milling or turning.

To conclude, the main shortcomings of the existing surface roughness models can be summarized as follows: (i) most of the developed models are restricted to FDM process and rarely consider other AM processes or hybrid scenarios; and (ii) the effect of secondary manufacturing process is not well addressed by the existing models. Therefore, the objective of this article is to address the above shortcomings by proposing novel mathematical models considering coupled operations of AM and milling process and a new surface profile representation scheme that can significantly reduce the prediction error.

3. Surface roughness modelling for an AM process

Among the existing surface roughness measures, the arithmetic average height parameter, R_a, is known to be the most widely used parameter and provides a good general description of height variations along the length of a specimen's surface (Gadelmawla *et al.*, 2002). It is theoretically defined as the arithmetic mean of the actual profile's departure from the mean line along the sampling length (Sander, 1991; ISO, 1997). The mathematical definition of R_a is as follows:

$$R_a = \frac{1}{l}\int_0^l |y(x)|dx, \tag{1}$$

where $y(x)$ is the surface profile function and l is the evaluation length of the profile.

The accuracy of the above model, however, relies heavily on the accuracy of the profile function $y(x)$. Two popular geometrical profile functions exist in the literature: (i) triangle function or staircase model (Luis Perez *et al.*, 2001); and (ii) semi-circular function (Boschetto *et al.*, 2012; Taufik and Jain, 2016), where the latter function is a more realistic representation of the surface profile. In addition, among the existing semi-circular functions in the literature, the parabolic function has been shown to lead to average higher prediction accuracies according to the experimental results (Taufik and Jain, 2016). The reason for this better performance lies in the fact that the parabolic function incorporates different error parameters along the layer base and height directions. In reality, the obtained layer thickness usually varies from the designed value, due to different errors caused by material, machine and process properties. Thus, by incorporating these errors into the surface profile function, the accuracy of the surface roughness model is expected to increase.

3.1. Surface profile representation

The proposed surface profile representation scheme is based on the combination of a parabolic curve and a linear line, as shown in Figure 1. The parabolic function is selected to incorporate the errors, and the linear function is added to provide a more accurate representation of the geometrical profile. The proposed surface profile is defined as a function of the stratification angle (θ) and layer thickness (t). The stratification angle (θ) refers to the angle between the normal vector of the surface of interest and the build direction. The layer thickness (t) represents the designed height of each layer, which is usually altered as a result of different errors during the printing process. To visualize the surface profile more clearly, a 2D (two-dimensional) sketch is shown in Figure 1 for different ranges of the stratification angle. When the stratification angle is equal to 90°, a special case occurs in which the surface profile is only represented using the parabolic function. In this work, the build direction is assumed to be vertical at all circumstances, which limits our work to three-axis AM processes.

3.2. Coordinate system illustration

Two different coordinate systems, XY and $X'Y'$, are considered for the modelling procedure, as shown in Figure 2. The $X'Y'$ coordinate system is established based on the shape of the surface. In the $X'Y'$ coordinate system, the surface profiles are periodic combinations of parabolic curve and linear line. This combination forms the unit geometry of the surface profile, which is repeated along the length of the surface. A second coordinate system (XY) is established for the unit geometry of the surface profile to further simplify the modelling procedure, as the integral function used for surface roughness calculation is not significantly affected by varying the coordinate system from $X'Y'$ to XY (as shown in Figure 2). The error generated as a result of this coordinate system change is taken to be negligible.

Another issue that arises by considering the parabolic function in the $X'Y'$ coordinate system is that more than one output might exist for a single x input that contrasts with the mathematical definition of function, which must be avoided. Note that in this model it is assumed that the mean profile line does not pass through the linear line element of the unit geometry.

3.3. Surface roughness model

The surface area of the unit geometry is shown in Figure 3(a). The non-periodic parabolic function of the unit geometry in the XY coordinate system can be formulated based on the values shown in Figure 3(b). Error coefficients of ε_x and ε_y are defined to represent the deviation of layer thickness along the X and Y directions, respectively. Therefore, the effect of different error sources, including geometry, process parameters, machine, and

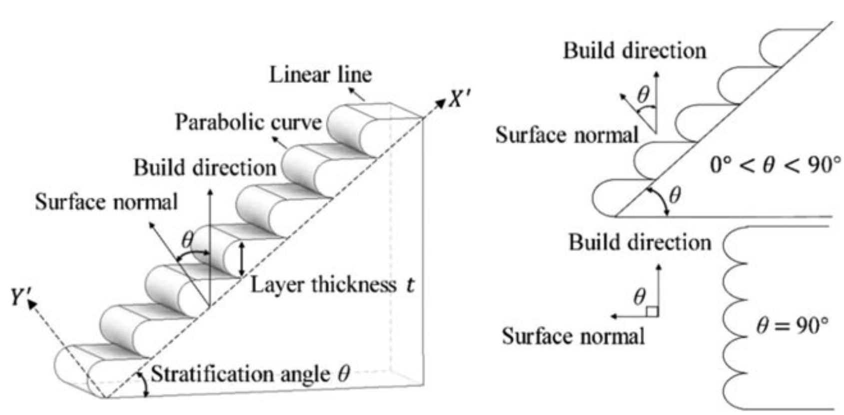

Figure 1. 3D and 2D illustrations of surface profile.

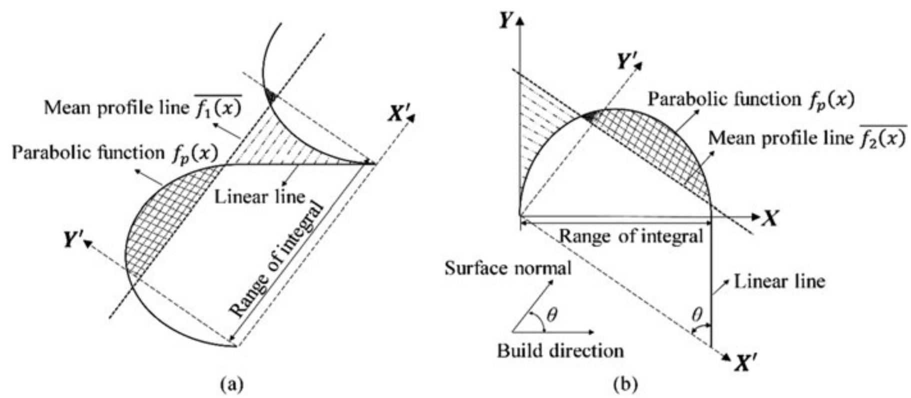

Figure 2. Illustration of unit geometry in (a) *X′Y′* and (b) *XY* coordinate systems.

even environment, can be incorporated into the model. Both error coefficients are assumed to follow normal distributions, where $\varepsilon_x \sim N(\mu_{\varepsilon_x}, \sigma^2_{\varepsilon_x})$ and $\varepsilon_y \sim N(\mu_{\varepsilon_y}, \sigma^2_{\varepsilon_y})$, in which their means and standard deviations are obtained by experiment. The non-periodic function of the parabolic curve $f_p(x)$ in the XY coordinate system can thus be represented by

$$f_p\,(x) = -2\left[\frac{t-2\varepsilon_y}{(t-\varepsilon_x)^2}\right]x^2 + 2\left[\frac{t-2\varepsilon_y}{(t-\varepsilon_x)}\right]x. \tag{2}$$

To find the mean profile line of the AM profile in XY coordinate system, the mean profile line in $X'Y'$ coordinate system $\overline{f_1(x)}$, as shown in Figure 3(a), is initially calculated using the following equation:

$$\overline{f_1\,(x)} = \frac{(S_p + S_t)}{(t-\varepsilon_x)}\sin\theta, \tag{3}$$

where S_p represents the surface area of the parabolic curve and S_t denotes the surface area of the triangle, as shown in Figure 3(a). Parameters S_p and S_t can be obtained from the following equations:

$$S_p = \frac{(t-\varepsilon_x)\left(t-2\varepsilon_y\right)}{3}, \tag{4}$$

$$S_t = \frac{(t-\varepsilon_x)^2\cot\theta}{2}. \tag{5}$$

Using Equations (3) to (5), the mean profile line in $X'Y'$ coordinate system $\overline{f_1(x)}$ can be calculated as

$$\overline{f_1(x)} = \frac{1}{6}[2\sin\theta(t-2\varepsilon_y) + 3\cos\theta(t-\varepsilon_x)]. \tag{6}$$

The function $\overline{f_1(x)}$ is then reformulated by converting the coordinate system to $X'Y'$, and is referred to as $\overline{f_2(x)}$ which is presented in Equation (7):

$$\overline{f_2(x)} = -(\cot\theta)x + \frac{1}{3}(t-2\varepsilon_y) + \frac{1}{2}(t-\varepsilon_x)\cot\theta. \tag{7}$$

Therefore, the surface roughness of additive manufactured surfaces can be calculated as

$$R_a = \frac{\sin\theta}{(t-\varepsilon_x)}\int_0^{(t-\varepsilon_x)} |f_p(x) - \overline{f_2(x)}|dx. \tag{8}$$

4. Surface roughness modelling for hybrid process

In this study, the subtractive process is considered for surface finishing, which aims to improve the surface roughness of 3D printed parts. To simplify the modelling process, the following terminologies are adopted in this article. The term "additive profile" is used to refer to the obtained surface profile from the AM process; "milling profile" is used to refer to the obtained surface profile from the flat end milling process on a non-3D printed surface; and "intersected profile" is used to refer to the obtained surface profile from the flat end milling process on a 3D printed surface.

4.1. Surface profile representation

To understand and model the intersected profile, different parameters such as cutting parameters, milling tool geometry,

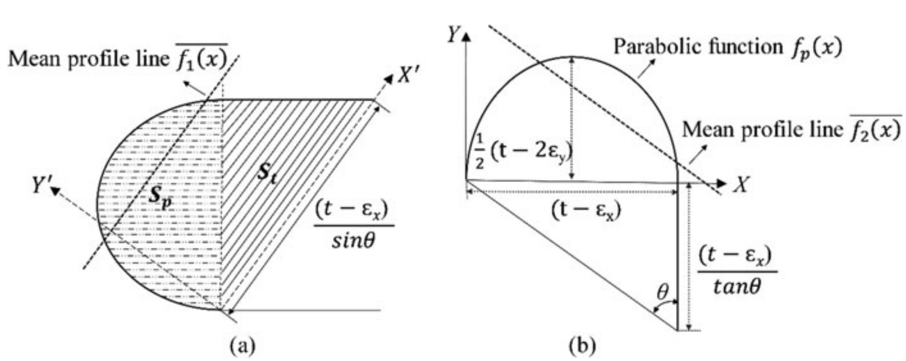

Figure 3. (a) Surface area of the unit geometry and (b) unit geometry parameters in *XY* coordinate system.

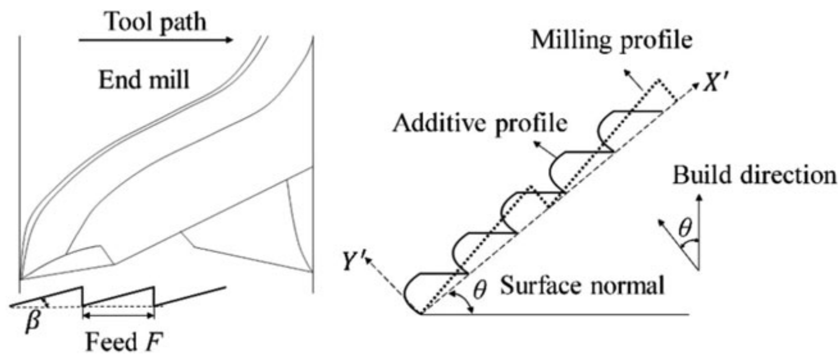

Figure 4. Representation of the milling profile.

and characteristics of the existing additive profile need to be considered. In this work, a sawtooth function is selected to represent the smallest unit of the milling profile in an ideal condition (Wang and Chang, 2004), as shown in Figure 4. Two parameters of feed (F) and working minor cutting edge angle (β) are incorporated into the sawtooth function to reflect the tool geometry and cutting parameters. The amplitude of the sawtooth function is defined as AP_s, where $A = \tan(\beta)$ is the slope of the sawtooth function and $P_s = F$ is the periodicity of the sawtooth function. Note that all the other parameters as well as the machine errors during the milling process are neglected in this work.

4.2. Surface roughness model

The following assumptions are adopted for the surface roughness modelling:

1. The starting coordinates of additive and milling profiles are similar.
2. The sawtooth function intersects with each parabolic curve at exactly two points.
3. The building direction is vertical at all circumstances.
4. The periodicity of sawtooth function (P_s) is always greater than the periodicity of parabolic function (P_p).
5. Both P_s and P_p are rational numbers.

To further analyze the intersected profile and surface roughness, two cases are considered based on the stratification angle: (I) $\theta = 90^\circ$ and (II) $0^\circ < \theta < 90^\circ$ as shown in Figure 5. Note that the special case $\theta = 0^\circ$ is not considered in this study.

4.2.1 Surface roughness model for $\theta = 90^\circ$

To study the obtained intersected profile, the periodic functions of both additive and milling profiles are derived using Fourier's theorem.

Fourier's theorem: It is known, that the general representation of Fourier trigonometric series for function $f(x)$ with a periodicity of L $(0 \le x \le L)$ is

$$f(x) = a_0 + \sum_{n=1}^{\infty}\left[a_n \cos\left(\frac{2\pi nx}{L}\right) + b_n \sin\left(\frac{2\pi nx}{L}\right)\right], \quad (9)$$

where coefficients a_0, a_n, and b_n can be obtained from Equations (10) to (12):

$$a_0 = \frac{1}{L}\int_0^L f(x)\,dx, \quad (10)$$

$$a_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{2\pi nx}{L}\right)dx, \quad (11)$$

$$b_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{2\pi nx}{L}\right)dx. \quad (12)$$

Using Fourier's theorem, Equations (9) to (12), the Fourier series representing the sawtooth function $f_s'(x)$ and periodic parabolic function $f_p'(x)$ can be derived as

$$f_s'(x) = \frac{AP_s}{2} + \sum_{n=1}^{\infty}\left[-\frac{AP_s}{\pi n}\sin\left(\frac{2\pi nx}{P_s}\right)\right], \quad (13)$$

$$f_p'(x) = \frac{1}{3}\left(t - 2\varepsilon_y\right) + \sum_{n=1}^{\infty}\left[-\frac{2\left(t - 2\varepsilon_y\right)}{\pi^2 n^2}\cos\left(\frac{2\pi nx}{P_p}\right)\right], \quad (14)$$

where P_p denotes the periodicity of parabolic function, and can be calculated as $P_p = (t - \varepsilon_x)$. The intersected profile $f_{p-s}'(x)$ can be calculated by subtracting the milling profile from the additive profile (shown in Figure 6), and can be represented

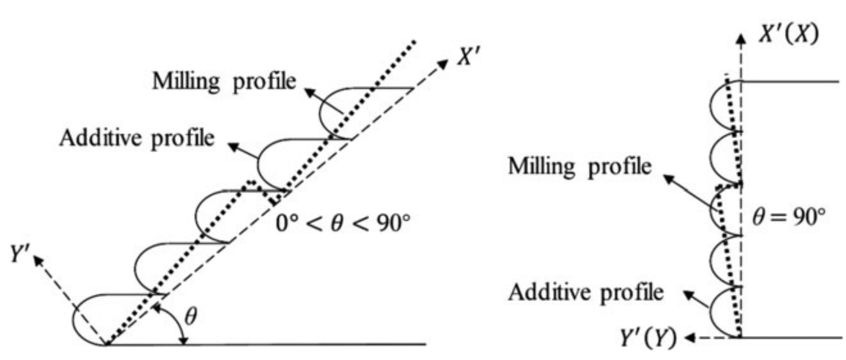

Figure 5. Representation of the intersected profile for different stratification angles.

using the following equation:

$$f'_{p-s}(x) = \min\{f'_s(x),\ f'_p(x)\}, \quad \forall x \tag{15}$$

The periodicity of the intersected profile can be calculated as $\psi = \text{LCM}(P_p, P_s)$, where LCM represents the least common multiple. Note that the periodicity of the intersected profile ψ always exists according to the model assumptions and Lemma 1:

Lemma 1. *If $x,\ y \in \mathbb{Q}$, then $LCM(x, y) \neq \emptyset$.*
Therefore, Equation (15) can be re-written as

$$f_{p-s}(x) = \min\{f'_s(x),\ f'_p(x)\}, \quad \forall x \in [0, \psi] \tag{16}$$

The mean of the intersected profile is shown in Figure 6, and can be calculated as shown in Equation (17):

$$\overline{f'_{p-s}(x)} = \frac{W}{\psi}, \tag{17}$$

where W denotes the surface area of the intersected profile, and can be obtained by the following equation:

$$W = \frac{\int_0^{\psi} f'_p(x)\,dx + \int_0^{\psi} f'_s(x)\,dx - \int_0^{\psi} \left|f'_p(x) - f'_s(x)\right| dx}{2}, \tag{18}$$

and $\int_0^{\psi} f'_p(x)dx$, $\int_0^{\psi} f'_s(x)dx$, and $\int_0^{\psi} |f'_p(x) - f'_s(x)|dx$ can be calculated using Equations (19) to (21):

$$\int_0^{\psi} f'_p(x)\,dx = \frac{\psi}{3}\left(t - 2\varepsilon_y\right) + \sum_{n=1}^{\infty}\left[-\frac{\left(t-2\varepsilon_y\right)\left(t-\varepsilon_x\right)}{\pi^3 n^3}\sin\left(\frac{2\pi n\psi}{t-\varepsilon_x}\right)\right], \tag{19}$$

$$\int_0^{\psi} f'_s(x)\,dx = \frac{\tan(\beta)F\psi}{2} + \sum_{n=1}^{\infty}\left[\frac{\tan(\beta)F^2}{2\pi^2 n^2}\cos\left(\frac{2\pi n\psi}{F}\right) - \frac{\tan(\beta)F^2}{2\pi^2 n^2}\right], \tag{20}$$

$$\int_0^{\psi}\left|f'_p(x) - f'_s(x)\right| dx = \int_0^{\psi}\left|\left(\frac{2\left(t-2\varepsilon_y\right) - 3\tan(\beta)F}{6}\right) + \sum_{n=1}^{\infty}\left[-\frac{2\left(t-2\varepsilon_y\right)}{\pi^2 n^2}\cos\left(\frac{2\pi nx}{t-\varepsilon_x}\right) + \frac{\tan(\beta)F}{\pi n}\sin\left(\frac{2\pi nx}{F}\right)\right]\right| dx. \tag{21}$$

Finally, the surface roughness of the intersected profile can be calculated as

$$R_a = \frac{1}{\psi}\int_0^{\psi}\left|f_{p-s}(x) - \overline{f_{p-s}(x)}\right|dx, \tag{22}$$

where $\overline{f_{p-s}(x)} = \overline{f'_{p-s}(x)}$, and can be calculated using Equation (17). Note that to solve the above integral, the $f_{p-s}(x)$ function, as in Equation (16), should be re-written, so that the value of the function is known for any given input, x. To do so, the intersection points between the parabolic and sawtooth functions for one period of ψ need to be determined. The $f_{p-s}(x)$ function in Equation (16) can thus be re-written as

$$f_{p-s}(x) = \begin{cases} f_s(x) & x_m^1 \le x < x_m^2 \\ f_p(x) & x_m^2 \le x < x_{m+1}^1 \end{cases} \qquad m = 0, \ldots, M-1. \tag{23}$$

where $x_m^{1,2}$ are the intersection points between the additive and milling profiles as shown in Figure 6, and can be calculated using the proposed algorithm in Section 5. Finally, using Equations (17) to (23), the surface roughness can be calculated as follows:

$$R_a = \frac{1}{\psi}\sum_{m=0}^{M-1}\left[\int_{x_m^1}^{x_m^2}\left|f_s(x) - \overline{f_{p-s}(x)}\right|dx + \int_{x_m^2}^{x_{m+1}^1}\left|f_p(x) - \overline{f_{p-s}(x)}\right|dx\right]. \tag{24}$$

4.2.2 Surface roughness model for $0^\circ < \theta < 90^\circ$

For cases where the stratification angle θ is within the range of $(0^\circ,\ 90^\circ)$, the surface profile is simplified to a linear function as shown in Figure 7. The reason for such an approximation is the unnecessarily long computation time for surface roughness calculation, which might not necessarily lead to a significant improvement of the prediction accuracy. Furthermore, since the amplitude of the sawtooth function is very small, and its periodicity is much larger than the additive profile, a linear function

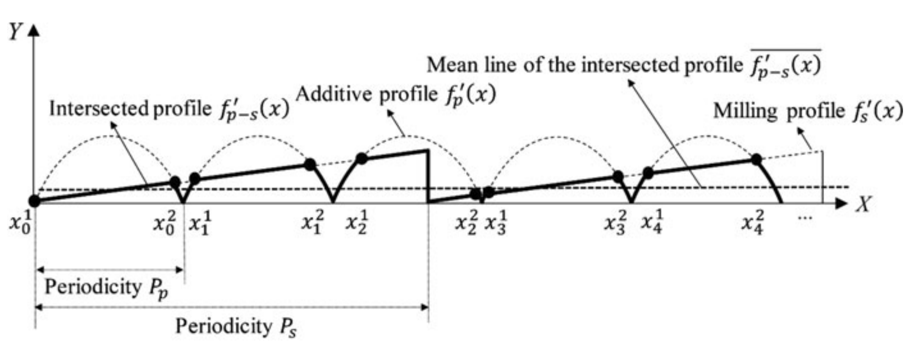

Figure 6. Illustration of the intersected profile and the intersection points.

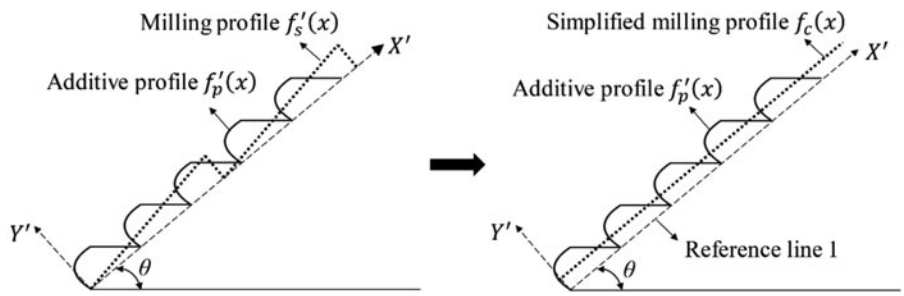

Figure 7. Illustration of the milling profile for $0^\circ < \theta < 90^\circ$.

can provide a good approximation. Therefore, for $0^\circ < \theta < 90^\circ$, the milling profile is represented using a linear function $f_c(x)$, which is parallel to the reference line 1 as shown in Figure 7.

Clearly, the axial depth of cut will highly affect the obtained profile, since the additive profile is represented using the combination of two different functions. As can be seen in Figure 8, any milling profile above the reference line 2, will only cut through the parabolic function. However, both the parabolic and linear functions will be involved in the intersected profile if the milling profile is lower than reference line 2. Therefore, two separate cases are considered. The distance between the reference lines 1 and 2 is $(t - K_b)\cos\theta$, and the distance between the milling profile position and reference line 1 is represented by D, as shown in Figure 8. Note that, in practical applications, the optimum axial depth of cut should be decided based on both the surface roughness and dimensional accuracy requirements. In addition, due to the existence of voids inside the printed part, increasing the depth of cut might not necessarily be beneficial.

Case 1: The milling profile is above or on reference line 2 ($D \geq (t - \varepsilon_x)\cos\theta$)

The mean line of the intersected profile $\overline{f_{p-s}(x)}$ in $X'Y'$ coordinate system can be calculated as

$$\overline{f_{p-s}(x)} = \frac{\sin\theta}{t-\varepsilon_x}\left[\frac{(t-\varepsilon_x)^2}{2\tan\theta} + \frac{(t-\varepsilon_x)\left(t-2\varepsilon_y\right)}{3} - \int_{x_0}^{x_1}\left(f_p(x) - f_c(x)\right)dx\right], \tag{25}$$

where the linear function $f_c(x)$ representing the milling profile can be formulated as

$$f_c(x) = -\cot\theta\left(x - \frac{D}{\cos\theta}\right), \tag{26}$$

and $f_p(x)$ is presented in Equation (2). Finally, the surface roughness can be calculated as

$$R_a = \frac{\sin\theta}{t-\varepsilon_x}\left[\int_0^{x_0}\left|f_p(x) - f_{p-s}^*(x)\right|dx + \int_{x_0}^{x_1}\left(f_c(x) - f_{p-s}^*(x)\right)dx + \int_{x_1}^{t-\varepsilon_x}\left(f_p(x) - f_{p-s}^*(x)\right)dx + \int_{x_1}^{t-\varepsilon_x}\left|-f_{p-s}^*(x)\right|dx\right], \tag{27}$$

where x_0 and x_1 are the intersection points of $f_p(x)$ and $f_c(x)$. Also, $f_{p-s}^*(x)$, the mean line in XY coordinate system can be obtained as shown in Equation (28):

$$f_{p-s}^*(x) = -\cot\theta\left(x - \frac{\overline{f_{p-s}(x)}}{\cos\theta}\right). \tag{28}$$

Case 2: The milling profile is below reference line 2 ($D < (t - \varepsilon_x)\cos\theta$)

In this case, both additive and milling profiles are involved in forming the intersected profile $f_{p-s}(x)$. Similarly, the mean of the intersected profile can be calculated as

$$\overline{f_{p-s}(x)} = \frac{\sin\theta}{t-\varepsilon_x}\left[\frac{(t-\varepsilon_x)^2}{2\tan\theta} + \frac{(t-\varepsilon_x)\left(t-2\varepsilon_y\right)}{3} - \frac{((t-\varepsilon_x)\cos\theta - D)^2}{\sin 2\theta} - \int_{x_0}^{x_1}\left(f_p(x) - f_c(x)\right)dx - \int_{x_1}^{t-\varepsilon_x} f_p(x)\,dx\right], \tag{29}$$

where x_0 is the intersection point of $f_p(x)$ and $f_c(x)$, and x_1 is the intersection point of $f_c(x)$ with the function $f(x) = 0$ in the XY

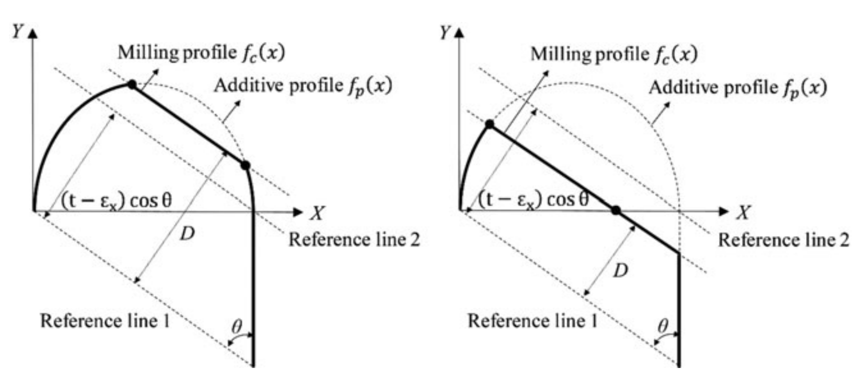

Figure 8. Different milling profile scenarios: case 1 (left) and case 2 (right).

coordinate system. Finally, using the mean line in XY coordinate system calculated by Equation (29), the surface roughness can be calculated as

$$R_a = \frac{\sin\theta}{t-\varepsilon_x}\left[\int_0^{x_0}\left|f_p(x) - f^*_{p-s}(x)\right|dx + \int_{x_0}^{t-\varepsilon_x}\left(f_c(x) - f^*_{p-s}(x)\right)dx\right]. \tag{30}$$

5. Solution algorithm for estimating the unknown parameters

The following algorithm is developed for determining the intersection points in Equation (23), which are necessary for calculating the surface roughness in Equation (24) when $\theta = 90^\circ$. In general, to calculate the intersection points of two periodic functions $f'(x)$ and $g'(x)$ with the maximum order of 2, the following is enforced:

$$f'(x) = g'(x). \tag{31}$$

However, if the periodic functions are represented with Fourier series, calculating the intersection points can become very complex. The following lemmas are thus used to simplify the calculation.

Lemma 2. *If $f'(x)$ is a periodic function:*

$$f'(x) = f'(x - \delta P_1), \quad \forall\delta \in \mathbb{Z},$$

where P_1 is the periodicity of function $f'(x)$.

Lemma 3. *If $f'(x)$ is a periodic function:*

$$f'(x) = f(x - \delta_1 P_1), \ \exists\delta_1 : \ x - \delta_1 P_1 \in [0, P_1] \text{ or } \delta_1 = \left\lfloor \frac{x}{P_1} \right\rfloor.$$

where $f(x)$ is the equivalent non-periodic function of $f'(x)$ for one period of P_1 and $\lfloor\,\rfloor$ is the floor function.

Using the above lemmas, Equation (31) is equivalent to

$$f(x - \delta_1 P_1) = g(x - \delta_2 P_2), \quad \forall x \in I, \tag{32}$$

where P_1 and P_2 are the periodicities of functions $f'(x)$ and $g'(x)$, respectively, δ_1 and δ_2 can be calculated based on Lemma 3, and I is the set of all intersection points.

Assuming that the periodicity of the intersected profile exists and is $P^* = LCM(P_1, P_2)$, it is known that:

$$\delta_1 \in \mathbb{Z} \cap \left\{0, \ \ldots, \frac{P^*}{P_1} - 1\right\}, \tag{33}$$

$$\delta_2 \in \mathbb{Z} \cap \left\{0, \ \ldots, \frac{P^*}{P_2} - 1\right\}. \tag{34}$$

Using all the feasible combinations of (δ_1, δ_2) from the above sets, where Equation (32) stands, the intersection points of the two functions can be calculated. However, first, we need to determine all of the feasible combinations of (δ_1, δ_2). For $\forall x \in I$ we can write:

$$\delta_1 P_1 + \alpha_1 = \delta_2 P_2 + \alpha_2, \tag{35}$$

where $0 \le \alpha_1 < P_1$ and $0 \le \alpha_2 < P_2$. Therefore, we have:

$$\delta_2 = \frac{\delta_1 P_1 + \alpha_1 - \alpha_2}{P_2}. \tag{36}$$

For any given $\delta_1 \in \mathbb{Z} \cap \{0, \ldots, \frac{P^*}{P_1} - 1\}$, we can thus write δ_2 as a function of the unknown value $(\alpha_1 - \alpha_2)$. However, the maximum and minimum values of $(\alpha_1 - \alpha_2)$ are known. Therefore, we have:

$$\delta_2 \in \mathbb{Z} \cap \left[\delta_1\left(\frac{P_1}{P_2}\right) - 1 \ , \ \ \delta_1\left(\frac{P_1}{P_2}\right) + \frac{P_1}{P_2}\right]. \tag{37}$$

Assuming that $P_1 < P_2$, we can derive that:

$$\delta_2 = \left\lfloor \left(\frac{P_1}{P_2}\right)(\delta_1 + 1) \right\rfloor. \tag{38}$$

Therefore, using Equations (35) to (38), all the feasible sets of (δ_1, δ_2) can be obtained, which are then used to calculate the intersection points. As we have assumed that the functions are of maximum order of two, we can get a maximum of two roots by solving the $f(x - \delta_1 P_1) = g(x - \delta_2 P_2)$ equation. Assuming that exactly two intersection points exist per each period P_1, the proposed algorithm can be summarized as follows:

Step 1. Set $\delta_1 = 0$ and $m = 0$.
Step 2. Set $\delta_2 = \lfloor(\frac{P_1}{P_2})(\delta_1 + 1)\rfloor$
Step 3. If the $f(x - \delta_1 P_1) = g(x - \delta_2 P_2)$ has any solution, go to **Step 4**.
Else set $\delta_1 = \delta_1 + 1$, and go to **Step 2**.
Step 4. Set $x^1_m = x_{min}$ (the smallest root) and set $x^2_m = x_{max}$ (the largest root).
Step 5. If $\delta_1 < \frac{P^*}{P_1}$, set $\delta_1 = \delta_1 + 1$ and $m = m + 1$, and go to **Step 2**.
Else, go to **Step 6**.
Step 6. End.

The set of all intersection points, I, can thus be written as

$$I = \left\{x^1_0, \ x^2_0, x^1_1, x^2_1, x^1_2, x^2_2, \ldots, x^1_M\right\} \text{ or}$$
$$I = \left\{x^{1,2}_m, \ x^1_M\right\} \quad m = 0, \ldots, (M-1), \tag{39}$$

where $x^1_M = P^*$. Also, note that the above set I is sequenced in a monotonically increasing order.

6. Model validation and case studies

In this section, the experimental design is introduced, and the proposed surface profile representation scheme is verified by using the obtained profile image and variation data. Furthermore, the surface roughness models for both the AM and hybrid processes are validated through a set of experiments and compared with the literature.

6.1. Experimental design

To validate the proposed model, a geometry containing a diverse range of stratification angles: 20°, 30°, 40°, 50°, 60°, 70°, and 90°, is designed as shown in Figure 9(a). To perform the case studies, a hybrid additive–subtractive process that has been designed and developed by the authors in their recent work is used (Li *et al.*, 2018), as shown in Figure 9(b). This hybrid process is built based on a six-axis robot arm, equipped with changeable end-effectors which can be adjusted to different additive and

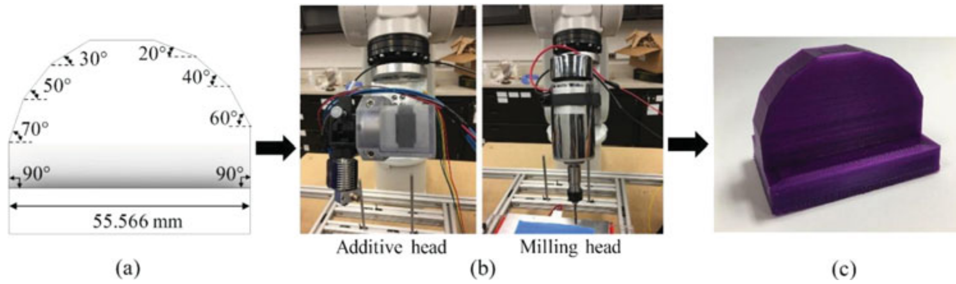

Figure 9. (a) Geometry design, (b) developed hybrid process, and (c) printed part.

subtractive processes. For the purpose of this work, two end-effectors are adopted, i.e., the FDM printing head and the milling head. The AM end-effector is first used to fabricate the test parts, which are then processed by the milling end-effector for better surface finish.

Polylactide (PLA) material is used in the FDM process for printing the test parts in a three-axis configuration (i.e., constant build direction). As frequently adopted in the literature, a layer thickness of 0.25 mm is selected to allow for the comparison of results. A four-flute end-mill tool with a diameter of 1/8 inch and nose radius of zero is then applied for further processing of the surfaces using a six-axis configuration (i.e., six degrees of freedom) which helps with reducing the setup time significantly.

Several replicated test parts are fabricated using both AM and the hybrid manufacturing process. Multiple surface roughness measurements are obtained for each surface (associated with each stratification angle) using the Bruker-Nano Contour GT-K Optical Profilometer. The surface height variation data are also obtained from the optical profilometer, which are then used for fitting the surface profile and comparison with the profile image. In addition, to evaluate the layer error coefficients and obtain a high-resolution image of the actual surface profiles, the Micro-Vu Precision Measurement System with a 1.0 µm scale resolution is used.

6.2. Surface profile analysis

To verify the proposed surface profile representation scheme, the two error coefficients (ε_x and ε_y) are first measured by the Micro-Vu System. The distributions of the error coefficients are obtained based on 10 different measurements for each surface profile and are presented in Table 1. The mean and standard deviation of each error element with respect to different stratification angles are then calculated based on the obtained measurements.

The profile variations ("raw profile data" shown in Figure 10(a)) are obtained using the Bruker-Nano Contour GT-K Optical Profilometer. These surface variation data are then further filtered using the smoothing spline function and used for fitting a quadratic function to each individual period/layer. The average fitted profile is illustrated in Figure 10(a) and is referred to as the "fitted profile." In addition, the actual surface profile image is obtained from the Micro-Vu System for comparison with the fitted profile and is presented in Figure 10(b). According to Figure 10(a) and (b), a periodic parabolic trend is observed in both fitted profile based on the raw data and the surface profile image, which verifies the adopted surface profile representation scheme used in the proposed surface roughness models.

6.3 Surface roughness model validation

Table 2 presents the results of the experiments to evaluate the effectiveness and validity of the proposed surface roughness models. A MATLAB code was developed and implemented for calculating the surface roughness values for both AM and hybrid cases based on the proposed mathematical models and the developed algorithm in which the experimental error coefficient values in Table 1 were considered. The main input parameters of the additive profile include the layer thickness (t), stratification angle (θ), and layer error coefficients (ε_x and ε_y); and for the milling profile include the working minor cutting edge angle (β) and feed (F). The milling profile parameters are obtained based on the milling tool geometry and the selected machining parameters including the robot's movement speed and the estimated spindle speed. In addition, the parameter n in Equations (13) and (14) is a main input parameter for the 90° hybrid case.

It is observed that the surface roughness generally tends to get smaller as the stratification angle increases, except for the 90° hybrid case. The average prediction errors for the AM and hybrid cases are found to be less than 5% and 9%, respectively. About a 21.35% error is observed for the 90° stratification angle in the hybrid case, which is higher than the errors obtained for

Table 1. Distributions of the error coefficients.

$\varepsilon_x \sim N(\mu_{\varepsilon_x}, \sigma^2_{\varepsilon_x})$	*Stratification angle θ (deg)*						
$\varepsilon_y \sim N(\mu_{\varepsilon_y}, \sigma^2_{\varepsilon_y})$	*20°*	*30°*	*40°*	*50°*	*60°*	*70°*	*90°*
μ_{ε_x} (mm)	0.0178	0.0102	0.0144	0.0114	0.0190	0.0421	0.0165
σ_{ε_x} (mm)	0.0144	0.0144	0.0205	0.0127	0.0173	0.0112	0.0149
μ_{ε_y} (mm)	0.0357	0.0483	0.0417	0.0541	0.0454	0.0817	0.0558
σ_{ε_y} (mm)	0.0162	0.0038	0.0063	0.0035	0.0043	0.0052	0.0073

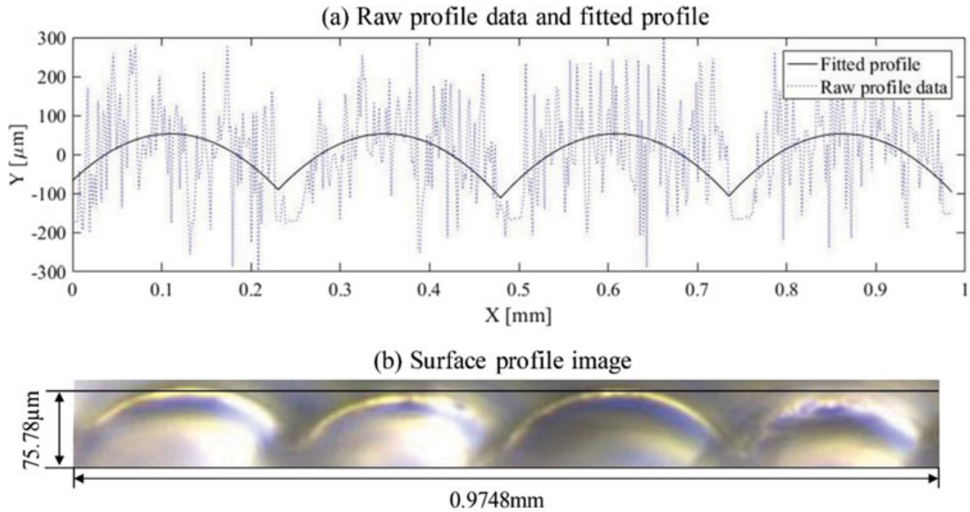

Figure 10. Comparison of (a) raw profile data and fitted profile, and (b) surface profile image.

other values of θ, due to a different model being used. In addition, the computation time and the accuracy of the sawtooth and periodic parabola functions in Equations (13) and (14), used in the 90° hybrid model, depend on the value of n. As n gets closer to infinity, the sine and cosine waves are further dampened, so that a more accurate approximation of the profile functions can be obtained. However, note that there is a difference between the accuracy of the proposed profile functions and the accuracy of the surface roughness prediction. Although ideally expected, a more accurate profile representation is not necessary associated with a more accurate prediction of surface roughness. Therefore, it is necessary to study the behavior of the model and the computation time to determine the optimum value of n. Therefore, the prediction accuracy is evaluated based on different values of n (2, 5, 50, and 100). Finally, $n = 2$ is selected as the best prediction accuracy is obtained. In addition to optimizing the model parameters, increasing the number of samples and measurements is expected to further reduce the obtained errors.

In Figure 11, the performance of both models is presented for different stratification angles. Confidence intervals are generated for the predicted roughness values based on the distribution of the error coefficients in Table 1. According to the figure, the confidence intervals are generally larger for smaller stratification angles. It should be noted that the confidence intervals for 70° and 90° are too narrow to be visible in the figure. The confidence intervals of additive and hybrid cases for the 70° angle are [19.312, 20.497] and [4.484, 5.379], and for 90° angle they are [16.426, 19.097] and [13.770, 14.007], respectively.

In Table 3, the accuracy of the proposed surface roughness model for the AM case is compared with the existing models

Table 2. Validation of the proposed models for AM and hybrid cases.

	AM case			*Hybrid case*			*% Ra reduction due to the secondary process*	
θ (deg)	*Observed Ra (μm)*	*Predicted Ra (μm)*	*Percentage error (%)*	*Observed Ra (μm)*	*Predicted Ra (μm)*	*Percentage error (%)*	*Observed (%)*	*Predicted (%)*
20°	54.069	55.023	1.764	24.511	25.562	4.288	54.667	53.543
30°	56.021	52.683	5.958	23.457	22.886	2.432	58.129	56.559
40°	50.651	46.789	7.625	15.340	17.159	11.862	69.715	63.327
50°	40.006	40.294	0.720	12.064	13.549	12.314	69.846	66.375
60°	30.808	32.560	5.689	8.065	7.768	3.683	73.821	76.143
70°	18.868	19.904	5.491	4.870	4.931	1.253	74.189	75.226
90°	17.332	17.762	2.484	11.445	13.889	21.354	33.966	21.805

Table 3. Comparison of the proposed AM surface roughness model with the literature.

Stratification	*Experimental*	*Pandey et al. (2003)*		*Boschetto and Bottini (2015)*		*Taufik and Jain (2016)*		*Proposed model*	
angle θ (deg)	*data (μm)*	*Prediction (μm)*	*Error (%)*	*Prediction (μm)*	*Error (%)*	*Prediction (μm)*	*Error (%)*	*Prediction (μm)*	*Error (%)*
0°	NA	28.150	NA	NA	NA	29.518	NA	NA	NA
20°	54.069	51.678	4.422	46.913	13.235	28.433	47.413	55.023	1.764
30°	56.021	35.377	36.850	32.090	42.717	33.465	40.263	52.683	5.958
40°	50.651	27.530	45.648	24.961	50.720	27.809	45.097	46.789	7.625
50°	40.006	23.105	42.246	20.943	47.650	23.334	41.674	40.294	0.720
60°	30.808	20.441	33.649	18.524	39.872	20.639	33.007	32.560	5.689
70°	18.868	18.840	0.148	17.071	9.524	19.019	0.800	19.904	5.491
90°	17.332	17.705	2.155	16.038	7.463	17.868	3.096	17.762	2.484

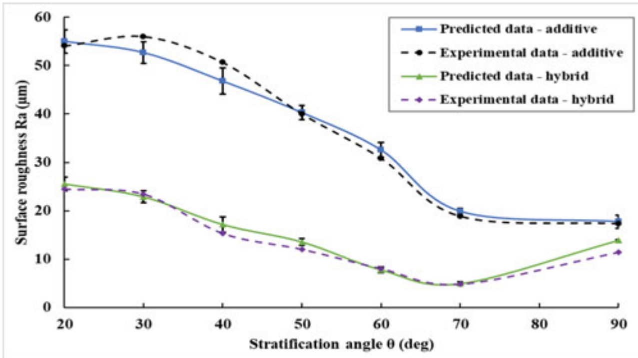

Figure 11. Comparison of predicted and experimental data in additive and hybrid cases.

in the literature. According to this comparison, the proposed model has superior performance compared with the existing models in most cases. The proposed representation scheme of the surface profile can be one of the reasons for this superior performance.

7. Conclusions and future work

In this article, mathematical models of surface roughness for both additive and hybrid additive–subtractive manufactured surfaces are established. Layer thickness, stratification angle, error coefficients, and parameters of the cutting tool are incorporated into the developed models. A new surface profile representation scheme is proposed to increase the prediction accuracy. Experiments are performed to verify the surface profile representation scheme and validate the developed models. According to the experimental results, the proposed models outperform the existing surface roughness models for the AM case, and also indicate promising results for the hybrid case. An average of 4.25% error is observed for the AM case, which is significantly smaller than the prediction error of the existing models in the literature. Furthermore, in the hybrid case, an average of 91.83% accuracy is obtained, which is significant specifically considering that no other similar models exist in the literature.

Extending the current model to the $\theta = 0^\circ$ or 180° scenario, considering other subtractive processes or profile models, and incorporating other parameters into the model are among the possible future works.

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